

4.3 Momentum

Mechanics · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Linear momentum $p=mv$ has direction; impulse $F\Delta t$ changes momentum. For collisions, total momentum is conserved if external impulse is negligible. Kinetic energy may decrease even when momentum is conserved, especially when bodies stick.

Worked example: follow the reasoning

A 2 kg mass moving 3 m s^{-1} sticks to a stationary 1 kg mass; shared speed is 2 m s^{-1} .

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Two 0.5 kg carts, one at 6 m s^{-1} and one stationary, stick. Find speed and lost kinetic energy. [3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

Momentum gives $v = (0.5 \times 6) / 1 = 3 \text{ m s}^{-1}$ [1]; KE initial 9 J [1], final 4.5 J, loss 4.5 J [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Mathematics 9709, 2026–2027. <https://www.cambridgeinternational.org/Images/697427-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.